

Luxury Stay Hospitality: Hotel Management System (HMS)

4th SEMESTER

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TABLE OF CONTENT

1. Executive Summary
2. Introduction
3. Problem Statement
4. Objective
5. Scope
6. Functional Requirements
7. Non Functional Requirements
8. System Arcitecture
9. Use case Diagram
10. Database Diagram
11. Module Description
12. User interface Screen
13. Security Features
14. Testing
15. Reports & Analytics
16. Future Enhacement
17. References
18. Appendix

Exclusive Summary.

Luxury Stay Hospitality aims to modernize hotel operations through a centralized Hotel Management System (HMS). The system automates reservations, room management, billing, housekeeping, guest services, and reporting to improve operational efficiency and guest satisfaction.

Introduction

The Hotel Management System is a web-based application developed to manage hotel operations efficiently. It provides functionality for staff, guests, and administrators to perform daily hotel activities through a centralized platform.

Problem Statement

Traditional hotel management processes are often manual, time-consuming, and prone to errors. Luxury Stay requires a digital solution that streamlines operations, improves customer experience, and supports strategic decision-making.

Objectives

- Automate hotel operations.
- Improve guest experience.
- Enhance room and reservation management.
- Generate accurate billing and invoices.
- Provide real-time reporting and analytics.
- Ensure security and data privacy.

Scope

The system covers:

- User Management
- Room Management.
- Reservations
- Check-in / Check-out
- Billing & Invoicing
- Housekeeping
- Maintenance Requests
- Feedback Management

Functional Requirments

User Management

- Admin Dashboard
- Staff Account Management
- Guest Profile Management

Room Management

- Room Inventory
- Room Booking
- Room Status Tracking

Reservation Management

- Online Booking
- Reservation Confirmation
- Booking Modifications

Check-In / Check-Out

- Room Allocation
- Billing Generation
- Status Updates

Billing & Invoicing

- Invoice Generation
- Tax Calculation
- Payment Recording

Housekeeping

- Cleaning Schedule
- Room Status Updates
- Maintenance Reporting

Reporting

- Occupancy Reports
- Revenue Reports
- Guest Feedback Reports

Non Functional Requirements

Performance:

- Response Time: 1-2 seconds
- Support for hundreds of concurrent users

Security:

- Encrypted Passwords
- Role-Based Access Control
- Secure Authentication

Reliability:

- 99% Uptime
- Automated Backups

Usability:

- Chrome
- Firefox
- Safari
- Edge
- Mobile Devices

System Architecture

- Presentation Layer
- Business Logic Layer
- Database Layer

Use Case Diagram

Actors:

- Admin
- Receptionist
- Housekeeping Staff
- Guest

Database Design

Tables:

- Users
- Guests
- Rooms
- Reservations
- Check-ins
- Invoices
- Payments
- Housekeeping Tasks
- Maintenance Requests
- Feedback

Module Description

Admin Module:

Manage users, settings, reports.

Reception Module:

Reservations, check-ins, check-outs.

Guest Module:

Bookings, feedback, service requests.

Housekeeping Module:

Cleaning schedules and room updates.

Maintenance Module:

Issue reporting and tracking.

User Interface Screens

- login Page
- Dashboard
- Room Management
- Reservation Page
- Guest Management
- Billing page
- Reporting Dashboard
- Feedback Page

Login Screen

LUXURYSTAY

Hospitality HMS

Welcome back to **LuxuryStay**

Manage rooms, bookings and hotel operations easily.

© 2026 LuxuryStay Hospitality

 Sign in

Login Portal

Enter your credentials to continue.

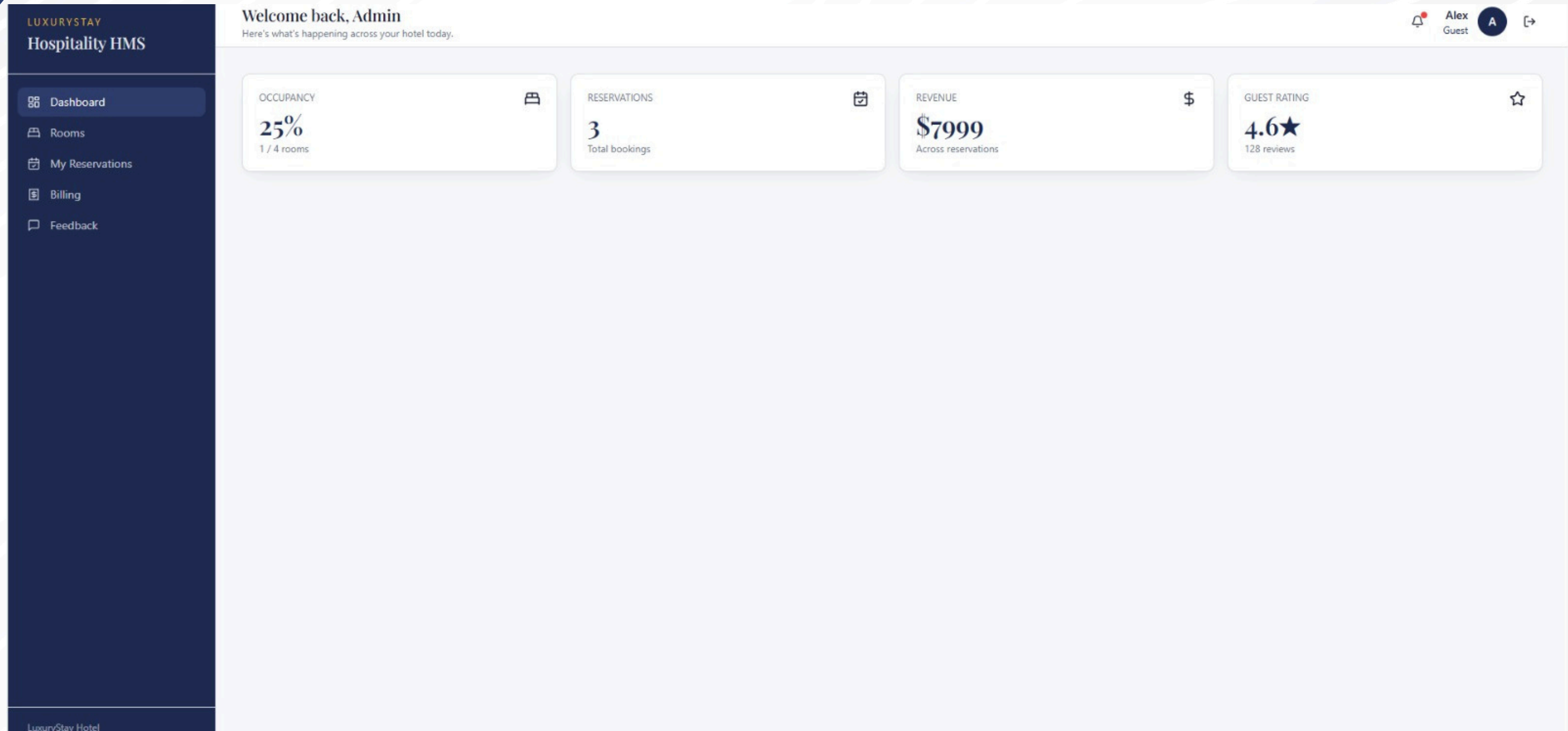
Email

Password

→ Sign in

New staff member? [Create an account](#)

Dashboard



Room Management

LUXURYSTAY
Hospitality HMS

Rooms
Live inventory across 5 floors and 40 rooms.

John Doe Admin

+ New Room

All Available Occupied Cleaning Maintenance

#101
Floor 2 - 4 guests
Deluxe
\$1222/night
Maintenance

#102
Floor 2 - 4 guests
Deluxe
\$5555/night
Available

Create New Room

Room ID

Floor

Guests

Price

Category
Standard

Status
Available

Cancel Create Room

Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Markup and share

LuxuryStay Hotel
v1.0 - static UI

Reservation Page

LUXURYSTAY
Hospitality HMS

Dashboard

Rooms

Reservations

Guests

Housekeeping

Billing

Reports

Staff

Feedback

Settings

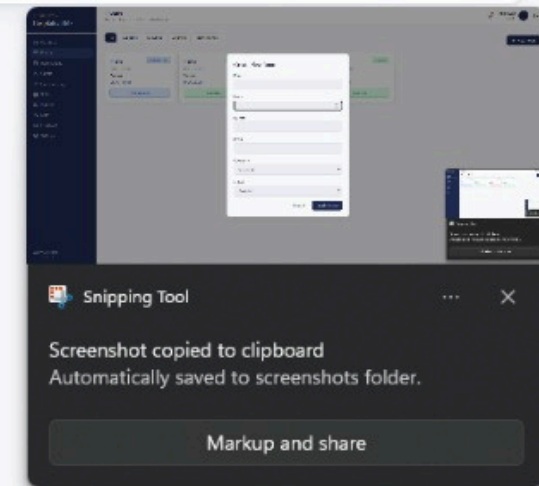
LuxuryStay Hotel
v1.0 - static UI

Reservations
All bookings

John Doe
Admin

+ New reservation

GUEST	ROOM	CHECK-IN	CHECK-OUT	STATUS	TOTAL	ACTION
Alex	#101 · deluxe	Wed Jun 10 2026	Sat Jun 13 2026	cancelled	\$ 1222	
Hamza	#102 · deluxe	Thu Jun 11 2026	Fri Jun 12 2026	completed	\$ 5555	Invoice Generated
Alex	#101 · deluxe	Wed Jun 10 2026	Sat Jun 13 2026	completed	\$ 1222	Invoice Generated
Ali	#102 · deluxe	Wed Jun 10 2026	Sat Jun 13 2026	completed	\$ 5555	Invoice Generated
Hamza	#101 · deluxe	Wed Jun 10 2026	Sat Jun 13 2026	completed	\$ 1222	Invoice Generated
Alex	#102 · deluxe	Wed Jun 10 2026	Fri Jun 19 2026	completed	\$ 5555	Invoice Generated
Hamza	#103 · standard	Fri Jun 19 2026	Sun Jun 21 2026	pending	\$ 3999	Confirm Cancel



Guest Management

LUXURYSTAY
Hospitality HMS

- Dashboard
- Rooms
- Reservations
- Guests**
- Housekeeping
- Billing
- Reports
- Staff
- Feedback
- Settings

LuxuryStay Hotel
v1.0 - static III
localhost:5173/reservations

Staff & Roles
Admin: manage team

John Doe Admin

NAME	EMAIL	PHONE	ROLE	STATUS	ACTIONS
Alex	alex@gmail.com	3122040000	Guest	Active	Edit
Hamza	hamza@gmail.com	312202040404	Guest	Active	Edit
Ali	ali@gmail.com	3122040279	Guest	Active	Edit

Snipping Tool

Screenshot copied to clipboard
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Markup and share

Billing Page

LUXURYSTAY
Hospitality HMS

Dashboard

Rooms

Reservations

Guests

Housekeeping

Billing

Reports

Staff

Feedback

Settings

LuxuryStay Hotel
v1.0 - static UI

Billing & Invoicing
Click invoice to view

John Doe
Admin

J

GUEST	ROOM	STATUS	TOTAL	ACTION
Alex	#102 · deluxe	PAID	\$49995	GET INVOICE
Hamza	#101 · deluxe	UNPAID	\$3666	GET INVOICE
Ali	#102 · deluxe	PAID	\$16665	GET INVOICE
Alex	#101 · deluxe	PAID	\$3666	GET INVOICE
Hamza	#102 · deluxe	PAID	\$5555	GET INVOICE

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Markup and share

Reporting Dashboard



Feedback Page

LUXURYSTAY
Hospitality HMS

Dashboard

Rooms

Reservations

Guests

Housekeeping

Billing

Reports

Staff

Feedback

Settings

LuxuryStay Hotel
v1.0 - static UI

Guest Feedback

Total reviews: 1

Alex

6/10/2026

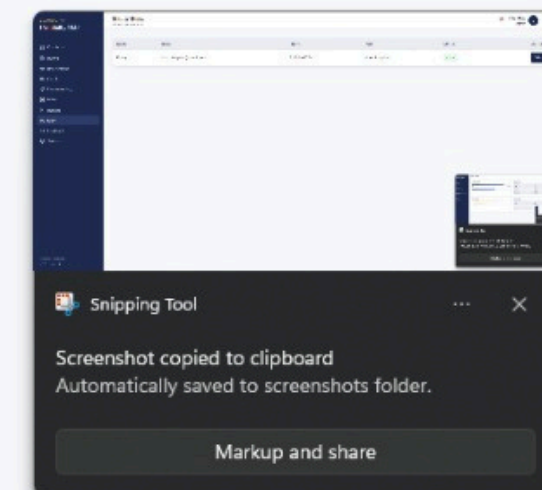
Service is good and on time

★ ★ ☆ ☆ ☆

John Doe

Admin

J



Security Features

- Password Hashing
- Role-Based Permissions
- Secure Session Management
- Data Encryption

Testing

- Unit Testing
- Integration Testing
- System Testing
- User Acceptance Testing

Reports & Analytics

- Occupancy Rate
- Revenue Analysis
- Guest Satisfaction
- Room Availability

Future Enhancements

- Mobile Application
- AI-Based Demand Forecasting
- Online Payment Gateway
- Chatbot Support

Conclusion

The Hotel Management System provides a complete solution for managing hotel operations efficiently. It improves service quality, automates processes, and enhances guest satisfaction.

References

- Software Engineering Books
- Database Management Concepts
- Web Development Documentation

THANK YOU